

A Transparent Quarterly Model of U.S. Fiscal Dynamics

Accounting, Treasury Refinancing, and Inflation

U.S. Fiscal Dynamics Simulator, version 0.1

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Abstract

This note describes a deterministic model of U.S. federal debt dynamics. The model is designed to answer accounting questions before attempting to model political or macroeconomic behavior. Its central feature is a security-level representation of marketable Treasury debt: fixed-rate securities retain their rates until maturity, bills roll over quickly, floating-rate notes reset, and Treasury Inflation-Protected Securities (TIPS) have inflation-indexed principal. Inflation, real growth, primary deficits, and new-issuance rates are independent assumptions. The resulting model is intentionally simple, but each result can be traced to an equation, an observed Treasury security, a CBO projection, or a stated approximation.

1 Introduction

A familiar fiscal projection shows federal debt relative to gross domestic product (GDP) rising for many years. Such a projection does not by itself explain how the debt stock would respond to a sudden change in inflation or interest rates. Two details are especially important. First, unexpected inflation can raise nominal GDP immediately, lowering the debt-to-GDP ratio even if no nominal debt disappears. Second, most Treasury debt does not instantly reprice: higher market rates enter federal interest costs gradually as securities mature, are refinanced, or reset.

Version 0.1 of the U.S. Fiscal Dynamics Simulator isolates those mechanisms. It is not a forecast, a general-equilibrium model, or a model of fiscal crisis. Congress, the Federal Reserve, and investors do not react endogenously. Instead, the user specifies paths for inflation, real growth, the primary deficit, other financing, and issuance rates. The model then calculates the implied GDP, interest costs, borrowing, refinancing, and debt stock.

The benchmark is the Congressional Budget Office’s (CBO’s) February 2026 baseline projection for 2026–2036 [1]. The starting Treasury portfolio is the September 30, 2025 Monthly Statement of the Public Debt (MSPD) [4].

2 Time, notation, and GDP

The simulation step is one calendar quarter. A federal fiscal year contains the December quarter of the preceding calendar year and the March, June, and September quarters of the named fiscal year. CBO budget flows are fiscal-year quantities and are allocated equally across those four quarters. CBO quarterly economic data retain their calendar-quarter timing.

Let Y_t denote nominal GDP at a seasonally adjusted annual rate, g_t annualized real GDP growth, and π_t annualized GDP-price inflation. Annual rates are converted to effective quarterly rates:

$$g_t^q = (1 + g_t)^{1/4} - 1, \quad \pi_t^q = (1 + \pi_t)^{1/4} - 1. \quad (1)$$

Nominal GDP evolves according to

$$Y_t = Y_{t-1}(1 + g_t^q)(1 + \pi_t^q). \quad (2)$$

This formulation makes inflation and real growth separate, visible assumptions. The model does not infer an effect of inflation on real output.

3 Fiscal and debt accounting

Let P_t be the primary deficit, with a positive value denoting a deficit and a negative value denoting a surplus. Let I_t be modeled debt interest and F_t the explicitly reported “other financing adjustment.” The modeled total deficit is

$$G_t = P_t + I_t, \quad (3)$$

and debt held by the public, D_t , satisfies

$$D_t = D_{t-1} + G_t + F_t. \quad (4)$$

The term F_t matters because changes in debt held by the public need not equal the unified budget deficit. Treasury cash-balance changes and federal credit-program cash flows are examples of other means of financing [1]. No hidden residual is used to force equation (4) to match a target series.

The treatment of TIPS makes it useful to distinguish interest expense from immediate cash borrowing. Write

$$I_t = C_t + I_t^O + X_t + L_t, \quad (5)$$

where C_t is marketable interest excluding TIPS principal indexation, I_t^O is the approximated interest cost of other public debt, X_t is TIPS inflation compensation, and L_t is any TIPS maturity-floor cost. TIPS indexation raises principal directly. Thus the signed cash financing requirement is

$$B_t = P_t + C_t + I_t^O + F_t, \quad (6)$$

while $X_t + L_t$ enters the debt stock through the TIPS liability itself. Consequently, $B_t + X_t + L_t = P_t + I_t + F_t$, preserving equation (4) without financing TIPS compensation twice. If $B_t < 0$, the model reports debt retirement rather than negative “new borrowing.”

Quarterly ratios use the annual-rate GDP denominator:

$$\left. \frac{P_t}{Y_t} \right|_{\text{annualized}} = \frac{4P_t}{Y_t}, \quad \left. \frac{I_t}{Y_t} \right|_{\text{annualized}} = \frac{4I_t}{Y_t}, \quad \frac{D_t}{Y_t} = \frac{D_t}{Y_t}. \quad (7)$$

4 Treasury cohorts and refinancing

Each marketable Treasury cohort i has a type k_i , issue quarter, maturity quarter, principal $Q_{i,t}$, fixed coupon, and effective financing rate $r_{i,t}$. The starting ledger contains 458 CUSIPs. At the end of quarter t , let $\mathcal{M}_t$ be the set of cohorts that mature. Maturing principal is

$$R_t = \sum_{i \in \mathcal{M}_t} Q_{i,t}. \quad (8)$$

The model removes R_t and issues the same amount as replacement debt. This operation changes the rate and maturity composition but not the total debt stock or the deficit. Gross Treasury issuance is therefore

$$\text{gross issuance}_t = R_t + \max(B_t, 0), \quad (9)$$

whereas genuinely new borrowing is only $\max(B_t, 0)$.

Bills and fixed-rate notes and bonds

Bills are carried at principal with a transparent effective-yield approximation. Their quarterly financing cost is $Q_{i,t}r_{i,t}/4$. Their short representative maturity makes bill costs respond rapidly to new issuance rates.

For notes and bonds, $r_{i,t}$ is fixed until maturity. A rate shock at time t therefore does not change the current cost of a long-dated inherited security. It affects that principal only after the security enters $\mathcal{M}_t$ and is replaced. This is the main source of gradual repricing in the model. Starting effective rates use issue-amount-weighted auction yields when available, otherwise coupons; premium and discount amortization are not modeled separately.

TIPS

For a TIPS cohort, let $\pi_t^{q,\text{ref}}$ be the quarterly reference-CPI change. The indexed principal and inflation compensation are

$$Q_{i,t}^+ = Q_{i,t}^-(1 + \pi_t^{q,\text{ref}}), \quad (10)$$

$$X_{i,t} = Q_{i,t}^+ - Q_{i,t}^-. \quad (11)$$

The real coupon is applied to indexed principal, so quarterly coupon cost is $Q_{i,t}^+ r_i / 4$. At maturity, redemption is the greater of indexed principal and original par. These conventions follow Treasury’s TIPS terms and federal budget treatment, under which principal adjustments and coupon accruals are recorded as interest [5, 2]. The official index is daily and uses a lagged reference CPI; the quarterly model approximates it with a one-quarter lag.

Floating-rate notes

Treasury floating-rate notes (FRNs) use the most recent 13-week bill auction rate plus a fixed auction spread and reset weekly [6]. At quarterly resolution, the simulator uses

$$r_{i,t}^{\text{FRN}} = r_t^{\text{short}} + s_i \quad (12)$$

and resets all FRNs once per quarter. FRN interest therefore responds to a rate shock without waiting for final maturity.

New issuance

New deficit borrowing is allocated using the starting public marketable composition:

Instrument	Issuance share	Representative tenor
Bills	21.54%	1 quarter
Notes	51.82%	7 years
Bonds	17.29%	20 years
TIPS	7.03%	10 years
FRNs	2.33%	2 years

This fixed mix is an explicit approximation, not a prediction of Treasury debt-management policy. Short, intermediate, long, TIPS-real, and FRN-spread assumptions are separately visible.

5 Debt coverage and interest definitions

The headline stock is federal debt held by the public, not gross federal debt. It is decomposed as

$$D_t = M_t + O_t, \quad (13)$$

where M_t is modeled marketable debt and O_t is other debt held by the public. MSPD security detail reports total amounts outstanding rather than public ownership by CUSIP. The model therefore scales CUSIP principal within each instrument class to the MSPD public-held class total. On September 30, 2025, the reconciliation is

$$\begin{aligned} D_0 &= \$30,277.766 \text{ billion,} \\ M_0 &= \$29,695.002 \text{ billion,} \\ O_0 &= \$582.764 \text{ billion.} \end{aligned}$$

Modeled debt interest is not relabeled as CBO net interest. CBO net interest consists primarily of interest on debt held by the public, offset by interest income and adjusted for other federal budget accounts [2]. The simulator retains CBO’s series as a comparison and reports the difference. CBO itself uses security-level principal, maturity, rate, financing needs, and issuance-mix assumptions when projecting net interest [3].

6 Scenarios and interpreting inflation

The baseline uses CBO paths for primary deficits, other financing, real GDP, inflation, 3-month bill rates, and 10-year note rates. Experimental scenarios may change inflation, real growth, the primary deficit, and issuance rates independently. This permits deliberately artificial comparisons—for example, high inflation with unchanged rates—to isolate accounting channels without pretending that such a combination is a forecast.

For a baseline b and scenario s , the debt-ratio difference is

$$\Delta d_t = \frac{D_t^s}{Y_t^s} - \frac{D_t^b}{Y_t^b}. \quad (14)$$

The simulator uses an exact two-factor Shapley decomposition. The nominal-debt contribution is

$$\Delta d_t^D = \frac{1}{2} \left[\left(\frac{D_t^s}{Y_t^b} - \frac{D_t^b}{Y_t^b} \right) + \left(\frac{D_t^s}{Y_t^s} - \frac{D_t^b}{Y_t^s} \right) \right], \quad (15)$$

and the nominal-GDP contribution is

$$\Delta d_t^Y = \frac{1}{2} \left[\left(\frac{D_t^b}{Y_t^s} - \frac{D_t^b}{Y_t^b} \right) + \left(\frac{D_t^s}{Y_t^s} - \frac{D_t^s}{Y_t^b} \right) \right]. \quad (16)$$

By construction, $\Delta d_t^D + \Delta d_t^Y = \Delta d_t$. This is accounting attribution, not a causal estimate. It makes the central interpretation explicit: inflation may lower debt/GDP through Y_t while nominal debt D_t continues to rise because of primary deficits, TIPS compensation, and refinancing costs.

7 Limitations

The model has no endogenous response of taxes, program spending, monetary policy, investor demand, exchange rates, or real economic activity. It does not model default, crisis probabilities, or a debt threshold. Federal Reserve holdings are included in debt held by the public, but Treasury and Federal Reserve balance sheets are not consolidated; reserve interest, remittances, and the Fed’s portfolio are outside the model. Other public debt has no security-level maturity structure, fiscal-year flows are spread evenly across quarters, and new issuance uses representative rather than auction-specific maturities. Results are therefore conditional accounting experiments and should always be reported together with their imposed inflation, rate, growth, and primary-deficit paths.

References

- [1] Congressional Budget Office. *The Budget and Economic Outlook: 2026 to 2036*. February 2026. <https://www.cbo.gov/publication/62105>.
- [2] Congressional Budget Office. *Federal Net Interest Costs: A Primer*. December 2020. <https://www.cbo.gov/publication/56910>.
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